

Products of sines and cosines - 3.2

Example: Find $\int \sin^8(x) \cos(x) dx$ $u = \sin(x)$
 $du = \cos(x) dx$

$$= \int u^8 du = \frac{u^9}{9} + C = \frac{\sin^9(x)}{9} + C$$

Something like this works when we have an odd power...

Example: Find $\int \sin^8(x) \cos^2(x) dx$

→ keep idea: $\sin^2(x) + \cos^2(x) = 1 \iff \cos^2(x) = 1 - \sin^2(x)$

$$= \int \sin^8(x) (1 - \sin^2(x)) \cos(x) dx = \int u^8 (1 - u^2) du$$

$$u = \sin(x) \quad du = \cos(x) dx$$

$$\int u^8 (1-u^2) du = \int u^8 - u^{10} du = \frac{u^9}{9} + \frac{u^{11}}{11} + C$$

$$= \frac{\sin^9(x)}{9} + \frac{\sin^{11}(x)}{11} + C.$$

How to handle larger powers:

$$\cos^7(x) = (\cos^2(x))^3 \cos(x) = (1 - \sin^2(x))^3 \cos(x)$$

Example: $\int \sin^5(x) \cos^3(x) dx = \int (1 - \cos^2(x))^2 \sin(x) \cos^2(x) dx$

$$u = \cos(x)$$

$$du = -\sin(x) dx$$

$$= -\int (1-u^2)^2 u^2 du = -\int (1 - 2u^2 + u^4) u^2 du$$

$$= \int -u^2 + 2u^4 - u^6 du = -\frac{u^3}{3} + \frac{2u^5}{5} - \frac{u^7}{7} + C$$

$$= \frac{-\cos^3(x)}{3} + \frac{2\cos^5(x)}{5} - \frac{\cos^7(x)}{7} + C.$$

Takeaway: we can integrate any product of powers of $\sin(x)$ and $\cos(x)$ as long as one of the powers is odd.

If both powers are even, use $\sin^2(x) = \frac{1 - \cos(2x)}{2}$ and $\cos^2(x) = \frac{1 + \cos(2x)}{2}$.

Example:
$$\int \sin^2(x) \cos^2(x) dx = \int \left(\frac{1 - \cos(2x)}{2} \right) \left(\frac{1 + \cos(2x)}{2} \right) dx$$
$$= \frac{1}{4} \int 1 - \cos^2(2x) dx = \frac{1}{4} \int \sin^2(2x) dx = \frac{1}{4} \int \frac{1 - \cos(4x)}{2} dx$$
$$= \frac{1}{8} \int dx - \frac{1}{8} \int \cos(4x) dx = \frac{1}{8} x - \frac{1}{32} \sin(4x) + C.$$